ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q3

Question 1

Given: sensor 1280 px wide, pixel pitch p = 5 µm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) (i) $f_x = f/p = 16/5 =$ **3.2 px**.

(ii) Width of sensor = 1280 × 5 = 6400. FOV = $N \cdot Z/f_x = 1280 \times 400 / 3.2 =$ **160000 mm**.

One pixel = FOV/N = 160000/1280 = **125 mm per pixel**. (looks large but the formula is right)

(b) Resolution of the ADC = $V_{range}/2^n = 5\text{ V} / 4096 = 1.22\times10^{-3}$ V = **1.22 mV**.

So the smallest change the system can resolve is 1.22 mV.

(c) With u held constant over [kT, (k+1)T] (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT}\,x(kT) + \int_0^T e^{A(T-s)}\,B\,u(kT)\,ds$. Since u(kT) is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma}\,B\,d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.

Substituting A = −2, B = 1, T = 0.1 s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.

Sanity check: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain $G/(1 - F) = 0.090635/0.18127 = 0.5$ ✓.

$$\mathbf{x_{k+1} = 0.8187\,x_k + 0.0906\,u_k}$$

Note: a longer lens would shrink the field of view in proportion (FOV ∝ 1/f).

Question 2

Given: J = 0.02 kg·m², b = 0.05 N·m·s/rad, m = 1 kg, ℓ = 0.25 m, g = 9.81 m/s². So mgℓ = 1 × 9.81 × 0.25 = 2.4525 N·m (used throughout).

(a) The model is second order, so I choose the position and velocity as states: $x = [x_1, x_2]^T = [\theta, \dot{\theta}]^T$, with input τ.

Rewriting $J\ddot{\theta} + b\dot{\theta} + mg\ell \sin\theta = \tau$ gives the nonlinear state-space model: $\dot{x}_1 = x_2$, $\dot{x}_2 = (1/J)(\tau - b\,x_2 - mg\ell \sin x_1)$, i.e. $\dot{x} = f(x, \tau)$.

Equilibria for constant τ_0 need $\dot{x} = 0$: the first equation gives $x_2 = 0$, and the second then gives $mg\ell \sin\theta_e = \tau_0$, i.e. $\sin\theta_e = \tau_0/(mg\ell)$.

This has a solution only if $|\tau_0/(mg\ell)| \le 1$, so $\mathbf{|\tau_0| \le mg\ell = 2.4525\ N\cdot m}$. For $|\tau_0| < mg\ell$ there are two equilibria per revolution (θ_e and $\pi - \theta_e$); at $|\tau_0| = mg\ell$ they merge, and beyond it the joint cannot be held statically.

(b) The equilibrium at $\theta_e = 60°$ needs the torque $\tau_e = mg\ell \sin 60° = 2.4525 \times 0.8660 = 2.124$ N·m. Define deviations $\delta x = x - x_e$ with $x_e = [60°, 0]^T$ and $\delta\tau = \tau - \tau_e$; then $\delta\dot{x} \approx A\,\delta x + B\,\delta\tau$ with the Jacobians evaluated at the equilibrium.

$\partial f_1/\partial x = [0, 1]$. $\partial f_2/\partial x_1 = -(mg\ell/J)\cos x_1 = -(2.4525/0.02)(0.5) = -122.625 \times 0.5 = -61.31$ (s^{-2}). $\partial f_2/\partial x_2 = -b/J = -0.05/0.02 = -2.5\ s^{-1}$.

A = [0, 1; −61.31, −2.5], and since $\partial f_2/\partial\tau = 1/J = 50$, **B = [0; 50]**.

Check: the characteristic equation $\lambda^2 + 2.5\lambda + 61.31 = 0$ gives $\lambda = -1.25 \pm 7.73j$, a lightly damped stable oscillation, as expected for a pendulum hanging near-vertical (60° below the downward vertical).

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$ so $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss}$. At θ_d the right side would be $2.4525 \neq 0$, so the error cannot be zero.

Estimate: $\theta_{ss} \approx \theta_d$ so $\sin\theta_{ss} \approx 1$; error $e \approx mg\ell/K_p = 2.4525/50 = 0.0491$ rad = **2.81°**.

(d) Steady state: $\dot{\theta} = \ddot{\theta} = 0$ and the control law gives $\tau = K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss}$. Plugging into the plant, $mg\ell \sin\theta_{ss} = K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss}$, so $K_p(\theta_d - \theta_{ss}) = 0$ and **$\theta_{ss} = \theta_d$: steady-state error is zero**.

Poles: linearise the closed loop about θ_d. Before compensation, $J\ddot{\theta} + (b + K_d)\dot{\theta} + (K_p + mg\ell \cos\theta_d)\,\delta\theta = 0$, characteristic polynomial $J\lambda^2 + (b + K_d)\lambda + K_p + mg\ell \cos\theta_d$. After compensation the added term $mg\ell \sin\theta$ cancels the plant's gravity term exactly, leaving $J\lambda^2 + (b + K_d)\lambda + K_p$.

The two differ only by $mg\ell \cos\theta_d$, and for $\theta_d = 90°$ this is $2.4525 \times 0 = 0$. So **the closed-loop poles are unchanged** at this operating angle. I should stress this is because $\cos 90° = 0$; if θ_d were, say, 60°, the compensaton would raise the stiffness by 61.3×0.02... i.e. it would move the poles.

Remark: the sag of a PD loop under gravity shrinks as K_p is increased.

Question 3

Given: A = [0, 1; 0, −2], B = [0; 1], C = [1, 0], f_s = 1 kHz so the sample time is $T = 1/f_s$ = 1 ms.

(a) Reason 1 – noise amplification: the backward difference $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T$ has frequency response $(1 - e^{-j\omega T})/T$, whose magnitude grows with frequency (≈ ω for small ωT), i.e. it is a high-pass operation. Any noice on the encoder signal therefore appears in the velocity multiplied by roughly $1/T = 1000\ s^{-1}$, and can swamp the true velocity.

Reason 2 – quantisation: the encoder only reports integer counts, so Δθ is an integer multiple of the count size q. Dividing by T = 1 ms gives velocity steps of q/T; for a 4096-count encoder $q = 2\pi/4096 = 1.53\times10^{-3}$ rad, so one step is 1.53 rad/s. At low speed the estimate flips between 0 and one step (a noisy staircase), exactly where the damping term of the PD law is needed.

(A further, smaller issue is the half-sample delay of a backward difference, but the two reasons above are the main ones.)

(b) $\omega_c = 2\pi \times 20 = 125.66$ rad/s; Tustin: $s = (2/T)(z - 1)/(z + 1)$; $a = \omega_c T/2 = 0.0628$.

$y_k =$ **$0.0591\,(u_k + u_{k-1}) + 0.8818\,y_{k-1}$**.

DC gain is unity because it is a low-pass filter.

(c) $A - LC = [-\ell_1, 1; -\ell_2, -2]$. $\det(sI - A + LC) = (s + \ell_1)(s + 2) - \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 - \ell_2)$.

Desiered $(s + 10)^2 = s^2 + 20s + 100$. So $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$ and $2\ell_1 - \ell_2 = 100 \Rightarrow \ell_2 = 36 - 100 = -64$.

$L = [18, -64]^T$.

(d) Advantage of the observer that no filter of this kind can give: the observer is driven by both the measurement y and the model with the actual input u, so it predicts the motion between samples. The velocity estimate therefore has little or no phase lag. The low-pass fliter on differenced data, by contrast, always pays a lag of about $1/(2\pi f_c) = 8$ ms for its noise rejection, a trade-off it cannot escape. In addition the observer gives all states, and can be augmented with a disturbance state (e.g. unknown load torque) which no filter on θ can estimate.

Circumstance where the filter is safer: when the plant model is poorly known or changes during operation, for example an unknown payload, varying friction, contact with the environment, or actuator saturation making u differ from the commanded value. The observer propagates the wrong model so its estimate can be biased (and in the worst case unstable with the feedback loop), whereas the model-free filter just smooths what the encoder actually measures.